

EMMANUEL SAMPSON

• ENGINEERING STUDENT • PHYSICS RESEARCHER

CONTACT

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EDUCATION

William & Mary
BS Engineering Physics and Applied Design
BA Philosophy with Honors
Expected Completion May 2026

UCLA School of Theater, Film & Television
Creative Production Program
Completed July 2024

Achievements

Winner of William & Mary 2024 and 2025 Hackathon

William and Mary Spirit Scholarship 2024
Selected as a student who exemplifies the William & Mary spirit: leadership, community and international service.

William and Mary Catron Scholarship
Awarded to students who represented great potential and ambition within the arts

William and Mary Bosanko Scholarship

Sharpe Community Scholar

Marine Officer Candidate School Attendant

KEY SKILLS

Python, HTML, SPSS

Jira Service Manager

Microsoft Office, Microsoft Excel

Fiber Optics & RF Engineering

CAD & 3D Modeling

Data Analysis & Signal Processing

Technical Writing

Computer Vision, Artificial Intelligence, and Machine Learning, Neural Networks

Conversational Japanese

Certifications

The Boehly Center for Excellence in Finance
Introduction To Python Training
Completed April 2024

From Excel to Tableau

Everybody's Introduction to Tableau (2022)

Tableau for Data Scientists

Corporate Finance: Environmental, Social, and Governance (ESG)

EXPERIENCE

ENGINEER INTERN

MAY 2025—PRESENT

WEINERT Fiber Optics Inc • Williamsburg, VA

Designed and implemented AI-powered computer vision systems in Python and C++ for automated fiber-optic end-face defect detection by building convolutional neural network with high performance computing clusters in order to develop a custom microscope and software program to automatically locate and identify surface defects on fiber-optic end-faces, thereby enhancing transmission efficiency for medical applications like kidney stone removal and cancer treatment.

PHYSICS RESEARCH ASSISTANT

OCT 2023—PRESENT

Applied Research Center Core Labs • Williamsburg, VA

Responsible for managing a \$ 4,000 budget for acquiring new electronics and equipment necessary for the lab. .

Assisted research teams by operating classical and electron microscopes and organizing experimental data.

Responsible for teaching and training graduate students and professionals on high-end material characteristic devices.

STUDENT MAKERSPACE ENGINEER

OCT 2022—NOV 2023

W&M Makerspace • Williamsburg, VA

Fulfilled over 1,000 service requests through the Microsoft Jira system for different unique machines such as Lazer Cutters, 3D Printers, and CNC Mills. Trained and certified over 200 students in the safe operation of advanced manufacturing equipment, providing expert advice for research and academic projects.

Designed and led a multi-level robotics workshop series, successfully accommodating participants from beginner to advanced skill levels

CAMPUS INVOLVEMENT AND EXTRACURRICULARS

Robotics Club President

Led robotics workshops for an average of 20 students per meeting from varying skill levels
Planned and Hosted the first William & Mary STEM Symposium by reaching out and connecting with 15 scientists and engineers who shared their stories with 150 William & Mary students.
Organized field trips and semester curriculum of projects that involved basic analog electronics, digital electronics, radio devices, and remote control.

Data Science Society President

Manages and hosts weekly workshops on aspects and tools for data science practices such as power bi, kaggle, and arcgis.
Developed additional workshops centering computer vision, machine learning, and neural networks.

Technical Executives Association President

Manages a collective group of 5 technical student organizations of Data Science, Computer Science, Artificial Intelligence, and additional subsets.

Tribe Innovation Marketing & Communications Lead

Aided in marketing efforts for a the consulting club, Tribe Innovation, which connects to external businesses and institutions to teach students about consulting practices.

Photography Club President

Coordinated and organized a schedule of student and faculty events
Developed and designed workshops focusing on the technical details as well as the social aspects of photography

Projects

Bird Sensor Hackathon 2024 Winner

In order to avoid window collisions, I created and pitched a device that would sense the electromagnetic distinctions of bird flight and their proximity to predict when they would hit windows to detour collisions

WaveFinderAI Hackathon 2025 Winner

By analyzing the unique frequencies reflected from microplastics in water, I simulated a tool that allowed the analysis of additional statistics such as the rate the water moves, the shapes and size of conaminants, and concentrations of contaminants for both machine hydraulics and other fluid based systems